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I confirm that I understand my coursework needs to be submitted online via MST Classroom under the relevant module page before the deadline for my assignment to be accepted and marked. I am fully aware that late submissions will be treated as non-submission and a mark of zero will be awarded.

Contents

1.Introduction

1.1 Explanation of the Topic and AI Concepts Used

Artificial Intelligence (AI) is a field of computer science that focuses on creating systems capable of performing tasks that normally require human intelligence, such as learning, reasoning, and prediction. AI systems are designed to mimic certain cognitive functions of humans by using computational models and algorithms. Over time, AI has evolved from simple rule-based systems to advanced data-driven approaches that can handle complex real-world problems.

One of the most important branches of AI is Machine Learning (ML), which enables systems to learn patterns from historical data and make predictions without being explicitly programmed. Instead of following fixed rules, machine learning models identify relationships within data and continuously improve their performance as more data becomes available. This makes machine learning particularly suitable for solving problems where manual rule creation is difficult or impractical.

Machine learning problems are commonly categorized into:

• **Regression** – predicting continuous values
• **Classification** – predicting discrete classes

This coursework focuses on a regression problem, where the goal is to predict a continuous numerical value. Regression models are widely used in real-world applications such as economic forecasting, healthcare analytics, and environmental modeling. In this study, regression techniques are applied to predict life expectancy, which is expressed as a continuous numerical outcome measured in years.

Supervised learning is used in this coursework, where the models are trained on labeled data containing both input features and known output values. This allows the algorithm to learn the relationship between explanatory variables and the target variable. Supervised regression is particularly effective when historical data is available, as is the case with global life expectancy indicators.

1.2 Introduction to the Problem Domain

Life expectancy is a widely used indicator to measure the overall health, economic stability, and quality of life of a country's population. It reflects the average number of years a newborn is expected to live under current mortality conditions. Life expectancy is commonly used by international organizations and policymakers to assess development progress and health system performance.

Life expectancy is influenced by multiple factors such as healthcare infrastructure, disease prevalence, mortality rates, education, and economic performance. Improvements in medical technology, vaccination programs, access to clean water, and education have contributed significantly to increased life expectancy across many regions of the world. Conversely, poor healthcare access, high disease burden, and economic instability often result in lower life expectancy.

Predicting life expectancy using traditional statistical approaches can be limited due to the complex and non-linear relationships between these factors. Many of the influencing variables interact with each other in ways that are difficult to capture using simple mathematical models. Therefore, machine learning techniques provide a more effective approach to model such complexity by learning patterns directly from data.

Machine learning models can capture both linear and non-linear relationships and can handle large datasets with multiple features. This makes them particularly suitable for analyzing global health data, where multiple indicators jointly influence outcomes such as life expectancy.

1.3 Scope of the Assignment

Through this coursework, the aim is to:

- Apply supervised machine learning algorithms
- Analyze health and economic indicators
- Compare multiple regression models
- Evaluate prediction performance using standard metrics

In addition to these objectives, the assignment also focuses on understanding the complete machine learning pipeline, from data collection to result interpretation. The coursework emphasizes practical implementation, data preprocessing, visualization, and model evaluation, ensuring a comprehensive understanding of applied machine learning concepts.

The scope of this work is limited to regression-based prediction using structured tabular data. Advanced techniques such as deep learning and time-series forecasting are outside the scope of this assignment but are identified as potential areas for future research.

2. Background

2.1 Research on Life Expectancy Prediction

Several studies have shown that factors such as healthcare expenditure, GDP per capita, disease prevalence, and mortality rates have a strong influence on life expectancy. Researchers have consistently identified healthcare accessibility and economic stability as major contributors to improved population health outcomes.

Previous research has applied regression-based models to predict life expectancy, with increasing interest in ensemble learning techniques. Linear regression has been commonly used due to its simplicity and interpretability. However, its assumption of linear relationships often limits predictive accuracy in complex real-world datasets.

Traditional models like Linear Regression are simple and interpretable but often fail to capture non-linear relationships. More advanced models such as Random Forest and Gradient Boosting overcome this limitation by combining multiple decision trees and learning complex interactions among variables. These models are particularly effective in handling noisy data and reducing overfitting.

2.2 Review of Existing Work

Existing works in this domain commonly use:

• Statistical regression techniques
• Tree-based machine learning models
• Ensemble learning approaches

Studies indicate that ensemble methods usually outperform single-model approaches in prediction accuracy. Random Forest and Gradient Boosting models are widely recognized for their robustness and ability to generalize well across diverse datasets. However, many existing works focus on only one algorithm, limiting comparative analysis.

This coursework addresses this gap by implementing and comparing multiple regression algorithms within the same experimental framework. This allows for a fair comparison of model performance and provides deeper insight into the strengths and limitations of each approach.

3. Dataset Background and Source

3.1 Dataset Source

The dataset used in this coursework was obtained from the **World Bank – World Development Indicators (WDI)** database.

Official Source:
• Organization: World Bank Group
• Database: World Development Indicators (WDI)
• Access Platform: World Bank DataBank

• Data URL:
https://databank.worldbank.org/source/world-development-indicators

The World Development Indicators database is a globally recognized and authoritative source that provides country-level data on health, education, economic performance, and social development. It is widely used in academic research, policy analysis, and international development studies due to its reliability and comprehensive coverage.

3.2 Dataset Selection Process

The dataset was not downloaded as a single predefined file. Instead, it was custom-built using the World Bank DataBank interface by selecting relevant indicators related to life expectancy and its influencing factors. This approach ensures transparency, originality, and academic integrity.

Selection Steps

The dataset was created using the following steps:

1. **Database Selection**
   o The World Development Indicators (WDI) database was selected from the World Bank DataBank.

2. **Country Selection**
   o All available countries were selected to ensure global representation and to avoid bias toward a specific region.

3. **Time Period Selection**
   o A multi-year range was selected (e.g., recent decades) to ensure sufficient historical data for analysis and model training.

4. **Indicator (Series) Selection**
   Indicators were selected based on their direct or indirect influence on life expectancy, supported by existing research in healthcare analytics.

3.3. Selected Indicators and Justification

The following indicators were selected from the WDI database:

Indicator	**Justification**
Life expectancy at birth, total (years)	Target variable to be predicted
Mortality rate, adult (male & female)	Reflects population health conditions

Indicator	Justification
Infant mortality rate	Indicator of healthcare quality
Under-5 mortality rate	Strongly linked to life expectancy
Diabetes prevalence	Represents chronic disease burden
Physicians (per 1,000 people)	Healthcare infrastructure indicator
Hospital beds (per 1,000 people)	Medical service availability
Current health expenditure (% of GDP)	National healthcare investment
GDP per capita (current US$)	Economic development indicator
School enrollment, secondary (% gross)	Education level and social development

These indicators were chosen because prior research has shown strong correlations between healthcare access, economic stability, education, and life expectancy.

Dataset Extraction and Format

After selecting the countries, indicators, and time period, the dataset was exported from the World Bank DataBank in CSV format.

The exported data initially contained:

- Country Name
- Country Code
- Indicator (Series) Name
- Indicator Code
- Year-wise values

Since the data was in a wide format, it was reshaped into a long format and then transformed into a machine-learning-friendly structure during preprocessing. This reshaping process was essential for applying regression algorithms efficiently.

3.4 Challenges and Limitations

- Presence of missing values in some indicators
- Variability across countries and years
- Need for proper feature scaling
- Risk of overfitting in complex models

Additional challenges included handling inconsistent reporting across countries and ensuring that sufficient data remained after cleaning for reliable model training.

4. Solution (Proposed Solution to the Chosen Problem)

4.1 Proposed Approach

The proposed solution uses a supervised machine learning pipeline to predict life expectancy. The solution consists of the following stages:

1. Data collection
2. Data preprocessing
3. Exploratory data analysis
4. Feature scaling
5. Model training
6. Model evaluation
7. Result visualization

Each stage is essential to ensure that the models are trained on clean, well-structured data and that the results are interpretable and reliable.

4.2 AI Algorithms Used

4.2.1 Linear Regression

Used as a baseline model to understand linear relationships between features and life expectancy. This model provides interpretability and serves as a reference point for evaluating more complex algorithms.

4.2.2 Random Forest Regressor

An ensemble learning method that builds multiple decision trees and averages their predictions to improve accuracy and reduce overfitting. Random Forest is particularly effective for handling non-linear relationships.

4.2.3 Gradient Boosting Regressor

A boosting-based ensemble method that builds models sequentially, correcting errors made by previous models. This method often achieves high predictive accuracy on structured datasets.

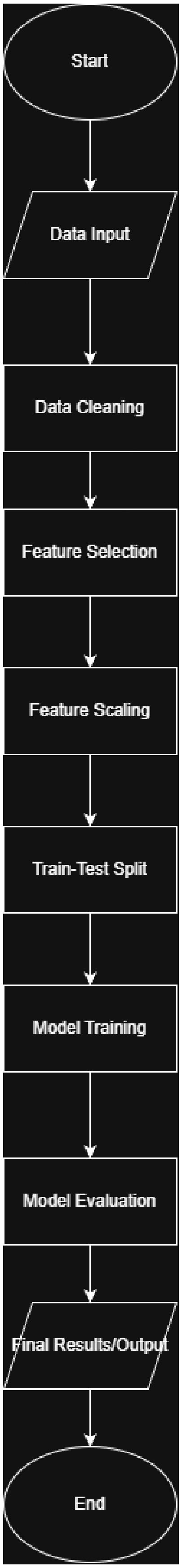
4.3 Pseudocode of the Solution

START
Import required libraries
Load dataset
Clean and preprocess data
Handle missing values
Scale numerical features
Split data into training and testing sets

Train Linear Regression model
Train Random Forest model
Train Gradient Boosting model

Evaluate models using MAE, MSE, and R^2 score
Visualize actual vs predicted values
END

4.4 Diagrammatic Representation


Start
Data Input
Data Cleaning
Feature Selection
Feature Scaling
Train-Test Split
Model Training
Model Evaluation
Final Results/Output
End

(This flowchart represents the complete lifecycle of the implemented solution.)

4.5. Explanation of the Development Process

(Tools, Technologies, and Libraries Used)

The AI prototype was developed using the Python programming language within a Jupyter Notebook environment. Jupyter Notebook was selected because it supports interactive development, allowing the code, outputs, and visualizations to be organized and executed step by step. This approach was useful for experimenting with different preprocessing techniques and machine learning models while clearly observing intermediate results.

The development process began with loading the dataset from a CSV file (500eccd6-936f-4340-b991-d197797ee9c9_Data.csv). The dataset contains multiple indicators recorded across different years. Since the raw data was initially structured in a wide format, it was reshaped into a long format to make it suitable for analysis and model training. During this stage, non-numeric values were handled and missing values were identified to ensure data consistency.

For data handling and preprocessing, the Pandas library was used extensively. Pandas enabled efficient data cleaning, reshaping using the melt operation, filtering of relevant indicators, and construction of a final structured dataset. NumPy was used to support numerical operations and array-based computations required during preprocessing and model evaluation.

Exploratory data analysis and visualization were carried out using Matplotlib and Seaborn. These libraries were used to generate plots such as distributions and correlation heatmaps, which helped in understanding relationships between features and identifying potential patterns or outliers in the data before training the models.

The machine learning implementation was performed using the Scikit-learn library. The prepared dataset was divided into training and testing subsets using train_test_split to allow objective evaluation of model performance. Feature scaling was applied using StandardScaler to normalize numerical features and improve learning performance for models sensitive to feature magnitude.

Multiple regression algorithms were implemented to predict the target variable. A Linear Regression model was first used as a baseline approach. To improve predictive performance and capture non-linear relationships, ensemble models such as Random Forest Regressor and Gradient Boosting Regressor were also trained. These models were chosen because they are robust to noise and can handle complex relationships in structured data.

Model performance was evaluated using standard regression metrics, including Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R^2 score. These metrics provided a quantitative measure of prediction accuracy and model effectiveness. The evaluation results were displayed directly within the notebook along with supporting visual outputs.

Overall, the development process followed a structured machine learning pipeline consisting of data preparation, feature processing, model training, evaluation, and result generation. This ensured that the developed AI application is functional, reproducible

5. Conclusion

5.1 Analysis of the Work Done

This coursework successfully demonstrates how machine learning techniques can be applied to a real-world healthcare problem. By comparing Linear Regression, Random Forest, and Gradient Boosting models, it was observed that ensemble methods provided better predictive performance.

5.2 Real-World Impact

Accurate life expectancy prediction can help governments and healthcare organizations:

• Identify critical health risk factors
• Allocate healthcare resources efficiently
• Improve long-term public health planning